PALM Installation, Configuration & Running a Simulation

Clone PALM model from `/storage` directory

```
> git clone /storage/data/palm/model.prod.git/ /home/gunes/palm/my.model.git
> git config --global --add safe.directory /storage/data/palm/model.prod.git//.git
```

Create build and `build/bin` directories manually

```
> cd palm/my.model.git/
> mkdir build
> mkdir build/bin
```

Create symlinks under `build/bin` directory

```
> ln -s /home/gunes/palm/my.model.git/bin/palmconfigure ./build/bin/palmconfigure
> ln -s /home/gunes/palm/my.model.git/bin/palmbuild ./build/bin/palmbuild
> ln -s /home/gunes/palm/my.model.git/bin/palmrun ./build/bin/palmrun
> ln -s /home/gunes/palm/my.model.git/bin/palmtest ./build/bin/palmtest
> ln -s /home/gunes/palm/my.model.git/bin/palmdoxygen ./build/bin/palmdoxygen
```

```
> ln -s /home/gunes/palm/my.model.git/bin/palm_cti ./build/bin/palm_cti
> ln -s /home/gunes/palm/my.model.git/bin/batch_scp ./build/bin/batch_scp
```

Always build and run within **`/build`** directory

```
> cd build
```

PALM Configuration

Create an MPI initialization script and source it to load the libraries

```
> nano palm_env.intel.sh
      # MPI initialization
      source /storage/libs/setenv_mpi_intel.sh
> source palm_env.intel.sh
```

Create a configuration file

```
> ./bin/palmconfigure –h
> ./bin/palmconfigure
```

Modify the configuration file

```
> nano .palm.config.default
```

```
%defaultqueue        qprod

%submit_command      qsub

%c_compiler_name     icx

%compiler_name       mpiifx

%compiler_name_ser   ifx

%cpp_options         -fpp -D__parallel -D__intel_compiler -DMPI_REAL=MPI_DOUBLE_PRECISION -
DMPI_2REAL=MPI_2DOUBLE_PRECISION -D__lc -D__netcdf -D__netcdf4_parallel -D__fftw -D__rrtmg

%make_options        -j 128

%compiler_options    -Ofast -xCASCADELAKE -mtune=cascadelake -g -traceback -fp-speculation=safe -ftz -no-
prec-div -fpe0 -r8 -no-wrap-margin -I${NETCDF_INCLUDE} -I${MKLROOT}/include -I${MKLROOT}/include/fftw -
I${RRTMG_INCLUDE}

%linker_options      -Ofast -xCASCADELAKE -mtune=cascadelake -g -traceback -fp-speculation=safe -ftz -no-
prec-div -fpe0 -r8 -no-wrap-margin -L${NETCDF_LIB} -lnetcdf -lnetcdff  -Wl,--start-group
${MKLROOT}/lib/libmkl_intel_ilp64.a ${MKLROOT}/lib/libmkl_sequential.a ${MKLROOT}/lib/libmkl_core.a -Wl,--
end-group -lpthread -lm -ldl  ${RRTMG_LIB}/librrtmg.so

%execute_command     mpirun -n {{mpi_tasks}} ./palm

%memory              4000

%login_init_cmd      . /home/gunes/palm/my.model.git/build/palm_env.intel.sh
```

Building the PALM model

```
> ./bin/palmbuild -c default
```

Get the INPUT files

```
> cp -rp /storage/data/palm/model.prod.git/build/JOBS/urban_environment/ ./JOBS/urban_environment
> rm -r ./JOBS/urban_environment/OUTPUT/
> rm -r ./JOBS/urban_environment/MONITORING/
```

Running the PALM model in batch mode

```
> ./bin/palmrun ?
>  nano urban_environment.sh
      # Run urban environment scenario
      ./bin/palmrun -b -k -r urban_environment -c default -a "d3# restart" -t $((48*3600)) -X 4 -T 4
-B -q qprod

> source urban_environment.sh
make: Nothing to be done for 'all'.
  *** executable and other sources created
```

```
  *** local jobcatalog "/home/gunes/palm/my.model.git/build/JOBS/urban_environment/LOG_FILES" does not exist and will be created now
  *** submit the job
11589.master
  --> palmrun finished


> qsub
> CTRL+D
    Job script will be read from standard input. Submit with CTRL+D.
    11590.master
> qstat
> qstat 11590.master -H
master:
                                                                  Req'd  Req'd   Elap
Job ID          Username Queue    Jobname    SessID NDS TSK Memory Time  S Time
--------------- -------- -------- ---------- ------ --- --- ------ ----- - -----
11590.master    gunes    qprod    STDIN         --    1   1    --  24:00 F 00:00
```

Running the PALM model interactively

```
> nano urban_environment.sh
      # No batch
      ./bin/palmrun -r urban_environment -c default -a "d3#" -t $((48*3600)) -X 4 -B

      # Run urban environment scenario: Job creation mode
      # ./bin/palmrun -b -k -r urban_environment -c default -a "d3# restart" -t $((48*3600)) -X 4 -T
      4 -B -q qprod
```

Modify the configuration file

```
> nano ./.palm.config.default
      #%defaultqueue        qprod
      #%submit_command      qsub
> source urban_environment.sh
*** execute command:
"mpirun -n 4 ./palm"
```

```
[1740730718.001064] [master:4113303:0]        ib_iface.c:1060 UCX  ERROR ibv_create_cq(cqe=4096) failed:
Cannot allocate memory : Please set max locked memory (ulimit -l) to 'unlimited' (current: 64 kbytes)
```

```
[1740730718.001097] [master:4113305:0]        ib_iface.c:1060 UCX  ERROR ibv_create_cq(cqe=4096) failed:
Cannot allocate memory : Please set max locked memory (ulimit -l) to 'unlimited' (current: 64 kbytes)

Abort(1614735) on node 0 (rank 0 in comm 0): Fatal error in PMPI_Init: Unknown error class, error stack:

MPIR_Init_thread(192)........:

MPID_Init(1665)..............:

MPIDI_OFI_mpi_init_hook(1665):

create_vni_context(2229).....: OFI endpoint open failed (ofi_init.c:2229:create_vni_context:Input/output
error)

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create_vni_context(2229).....: OFI endpoint open failed (ofi_init.c:2229:create_vni_context:Input/output
error)


  +++ runtime error occured

  *** Execution of ERROR-command:

  >>> [[ normal = execution ]] && cat RUN_CONTROL



+++ palmrun crashed
```